

Mini Study Guide 3: Mutual Fund Costs

Loads, 12b-1 Fees, and Turnover (BKM Ch. 4)

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Learning objectives

Fund costs compound. A 1% difference in annual fees can cost more than a 5% up-front load over a long holding period, and turnover costs never appear in the expense ratio.

After this guide you should be able to:

1. Name each fee a mutual fund investor can pay, and say when it is charged.
2. Compute NAV, offering price, and a fund's holding-period return, with and without a load.
3. Compare share classes over different horizons and find the break-even holding period.
4. Estimate the hidden cost of portfolio turnover.

Core concept: the fee menu

Cost	When charged	Typical size	In the expense ratio?
Front-end load	At purchase	Up to 8.5% by rule; usually 3–5% for stock funds	No
Back-end load (CDSC)	At redemption; declines each year held	Often starts at 5–6%, falls to 0 over about 6 years	No
Operating expenses (management fee, admin)	Continuously, deducted from assets	0.03% (index) to 1%+ (active) per year	Yes
12b-1 fee	Continuously, deducted from assets	Up to 1% per year (0.75% distribution + 0.25% service)	Yes
Trading costs from turnover	Whenever the fund trades	Depends on turnover and spreads	No , hidden

12b-1 fees pay for marketing, distribution, and broker compensation. They are named for the SEC rule that allows them. They are part of the expense ratio.

Share classes give the same portfolio different fee structures:

- **Class A:** front-end load, low 12b-1 fee. Best for long horizons.
- **Class B:** back-end load (CDSC), higher 12b-1 fee; often converts to Class A after several years. Now rare.
- **Class C:** little or no load, high 12b-1 fee (often 1%) every year. Best for short horizons.

Turnover is the share of the portfolio replaced each year. Commissions and bid-ask spreads reduce returns but are not in the expense ratio. High turnover also realizes capital gains, which taxable investors pay tax on each year.

Formulas

$$\text{NAV} = \frac{\text{Market value of assets} - \text{Liabilities}}{\text{Shares outstanding}}$$

Where:

- NAV = net asset value per share
- Market value of assets = current value of all securities and cash the fund holds
- Liabilities = what the fund owes (accrued fees, borrowings)
- Shares outstanding = number of fund shares held by investors

$$\text{Offering price} = \frac{\text{NAV}}{1 - \text{Front-end load}}$$

Where:

- Offering price = price per share an investor pays, including the load
- NAV = net asset value per share
- Front-end load = sales charge, as a decimal of the offering price

$$\text{Fund return} = \frac{\text{NAV}_1 - \text{NAV}_0 + \text{Income and capital gain distributions}}{\text{NAV}_0}$$

Where:

- NAV₀ = net asset value per share at the start of the period
- NAV₁ = net asset value per share at the end of the period
- Income and capital gain distributions = cash paid per share during the period

For an investor who paid a load, replace NAV₀ in both places with the offering price actually paid.

Fee drag over T years, with gross return r , expense ratio e , and front-end load L :

$$\text{Ending value} = \text{Investment} \times (1 - L) \times (1 + r - e)^T$$

Where:

- Investment = dollar amount invested
- L = front-end load, as a decimal
- r = annual return on the portfolio before expenses
- e = annual expense ratio, as a decimal (includes any 12b-1 fee)
- T = number of years held

Worked examples

Example 1: Fund return with and without a load (Algorithmic)

A fund starts the year at a NAV of \$20.00 and ends at \$21.10. It distributes \$0.50 of income and \$0.40 of capital gains. It has a 4% front-end load.

1. Return at NAV (no-load investor) = $(21.10 - 20.00 + 0.90) / 20.00 = \mathbf{10.0\%}$.
2. Offering price = $20.00 / (1 - 0.04) = \mathbf{\$20.83}$.
3. Return to the load-paying investor = $(21.10 + 0.90 - 20.83) / 20.83 = \mathbf{5.6\%}$.

The 4% load cost this investor 4.4 percentage points of first-year return.

Example 2: Share classes over time (Algorithmic)

You invest \$10,000. The portfolio earns 8% per year before expenses. Compare:

- **Class A:** 5% front-end load, 0.60% expense ratio. Grows at 7.40% on \$9,500.
- **Class C:** no load, 1.25% expense ratio (includes a 1.00% 12b-1 fee). Grows at 6.75% on \$10,000.

- **Index fund:** no load, 0.10% expense ratio. Grows at 7.90% on \$10,000.

Years held	Class A	Class C	Index fund
1	\$10,203	\$10,675	\$10,790
5	\$13,575	\$13,862	\$14,625
10	\$19,398	\$19,217	\$21,390
20	\$39,610	\$36,928	\$45,754

Break-even holding period (A vs. C): set $9,500 \times 1.074^T = 10,000 \times 1.0675^T$.

$$T = \frac{\ln(10,000/9,500)}{\ln(1.074/1.0675)} = \frac{0.05129}{0.00607} \approx 8.4 \text{ years}$$

Where:

- T = break-even holding period, in years
- 10,000 = amount invested in Class C (no load); 9,500 = amount invested in Class A after the 5% load
- 1.074 and 1.0675 = annual growth factors ($1 + r - e$) for Class A and Class C
- ln = natural logarithm

Hold less than about 8.4 years and Class C wins. Hold longer and Class A wins. The low-cost index fund wins at every horizon.

BA II Plus (Class A, 10 years): $10 \text{ N} \cdot 7.4 \text{ I/Y} \cdot 9500 \text{ +/- PV} \cdot 0 \text{ PMT} \cdot \text{CPT FV} \rightarrow 19,398$

Example 3: The hidden cost of turnover (Algorithmic)

A \$500 million fund has 80% annual turnover. Each round trip (sell one stock, buy another) costs 0.50% of the value traded in commissions and spreads.

1. Value traded = $0.80 \times \$500\text{M} = \400M .
2. Trading cost = $0.0050 \times \$400\text{M} = \2M .
3. As a share of assets: $\$2\text{M} / \$500\text{M} = \mathbf{0.40\% \text{ per year}}$.

If the fund's expense ratio is 0.90%, the investor's total annual cost is about 1.30%, but only 0.90% shows in the prospectus fee table.

Common pitfalls

- **Offering price.** It is $\text{NAV} / (1 - \text{load})$, not $\text{NAV} \times (1 + \text{load})$. The load is a percentage of the price you pay.
- **Leaving out distributions.** Fund return includes income and capital gain distributions, not just the NAV change.
- **"No-load" means "no fees."** No-load funds still charge expense ratios, and some charge 12b-1 fees up to 0.25%.
- **Assuming the expense ratio is the full cost.** Trading costs from turnover and the taxes on realized gains are extra.
- **Picking a share class without a horizon.** Class A vs. Class C depends on how long you will hold.

Key terms and definitions

- **Net asset value (NAV):** a fund's assets minus liabilities, divided by shares outstanding.
- **Offering price:** the price an investor pays per share; NAV plus any front-end load.

- **Front-end load:** a sales charge paid when shares are bought.
- **Back-end load / contingent deferred sales charge (CDSC):** a sales charge paid when shares are sold, declining the longer they are held.
- **12b-1 fees:** annual fees, up to 1% of assets, that pay for marketing, distribution, and shareholder servicing.
- **Expense ratio:** total annual operating expenses (management fees, administration, 12b-1 fees) as a percentage of assets.
- **Share classes (A, B, C):** different fee structures on the same portfolio: A = front load and low annual fees; B = back load; C = little or no load and high annual 12b-1 fees.
- **No-load fund:** a fund with no sales charge on purchase or redemption.
- **Portfolio turnover:** the share of the portfolio replaced during a year; drives hidden trading costs.
- **Soft dollars:** brokerage commissions that a fund pays partly in research and services from the broker, which can hide costs.
- **Open-end vs. closed-end fund:** open-end funds issue and redeem shares at NAV; closed-end funds have a fixed number of shares that trade on an exchange at a premium or discount to NAV.
- **Pass-through (tax) treatment:** a fund that distributes nearly all its income is not taxed itself; shareholders pay the tax.

CFA-style practice questions

Q1 [Algorithmic]. A fund holds \$250 million of assets and has \$10 million of liabilities and 12 million shares outstanding. Its NAV is *closest* to:

- A. \$20.00
- B. \$20.83
- C. \$21.67

Q2 [Algorithmic]. A fund has a NAV of \$19.00 and a 5% front-end load. Its offering price is *closest* to:

- A. \$18.05
- B. \$19.95
- C. \$20.00

Q3 [Algorithmic]. An investor puts \$10,000 into a fund with a 4% front-end load. The portfolio earns 9% before expenses, and the expense ratio is 1%. After one year the investment is worth *closest* to:

- A. \$10,368
- B. \$10,464
- C. \$10,800

Q4 [Conceptual]. 12b-1 fees are used *primarily* to pay for:

- A. the portfolio manager's performance bonus.
- B. marketing, distribution, and shareholder servicing.
- C. brokerage commissions on the fund's trades.

Q5 [Conceptual]. Which statement about trading costs from portfolio turnover is *most* accurate? They:

- A. are included in the fund's expense ratio.
- B. reduce the fund's returns but are not in the expense ratio.
- C. are billed to shareholders when they redeem.

Q6 [Algorithmic]. Fund X charges a 5% front-end load and a 0.50% expense ratio. Fund Y has no load and a 1.25% expense ratio. Both portfolios earn 8% before expenses. For a \$10,000 investment held 10 years:

- A. Fund X ends higher, at about \$19,580.
- B. Fund Y ends higher, at about \$19,217.
- C. Both end at about the same value.

Q7 [Algorithmic]. A no-load fund's NAV rises from \$30.00 to \$31.50, and it distributes \$1.20 per share. The rate of return is *closest* to:

- A. 4.0%
- B. 5.0%
- C. 9.0%

Q8 [Conceptual]. A contingent deferred sales charge (back-end load) typically:

- A. rises the longer shares are held.
- B. falls the longer shares are held.
- C. is charged when shares are bought.

Q9 [Algorithmic]. A fund has 60% annual turnover, and each round-trip trade costs 0.60% of the value traded. The annual trading cost as a percentage of fund assets is *closest* to:

- A. 0.36%
- B. 0.60%
- C. 1.00%

Answer key

Q	Answer	Explanation
1	A	NAV = $(250 - 10) / 12 = \$20.00$. C adds liabilities instead of subtracting them.
2	C	$19.00 / 0.95 = \$20.00$. B uses 19×1.05 ; A subtracts the load from NAV.
3	A	$\$9,600 \text{ invested} \times (1 + 0.09 - 0.01) = \$10,368$. B ignores the expense ratio; C ignores both costs.
4	B	12b-1 fees fund distribution and servicing. Commissions (C) are trading costs, not 12b-1 fees.
5	B	Trading costs reduce NAV but are not in the expense ratio.
6	A	X: $9,500 \times 1.075^{10} = \$19,580$. Y: $10,000 \times 1.0675^{10} = \$19,217$. The break-even is about 7.3 years, so X wins at 10 years.
7	C	$(31.50 - 30.00 + 1.20) / 30.00 = 9.0\%$. B leaves out the distribution.
8	B	The CDSC declines each year the shares are held, often to zero after about 6 years.
9	A	$0.60 \times 0.60\% = 0.36\%$ of assets per year.